# Shifan Xu

Ph.D. Candidate in Applied Physics

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## Education

**Yale University** — *New Haven, CT, USA*
Ph.D. Candidate in Applied Physics — *Sep. 2021 – Present*
Advisors: Prof. Yongshan Ding and Prof. Steven M. Girvin

**University of Chicago** — *Chicago, IL, USA*
Visiting Ph.D. Student; Host: Prof. Liang Jiang — *Jan. 2024 – May 2024*

**Nanjing University** — *Nanjing, China*
B.S. in Physics — *Sep. 2017 – Jun. 2021*

**University of California, Berkeley** — *Berkeley, CA, USA*
Visiting Undergraduate Student — *Jul. 2019 – Jan. 2020*

## Research Experience

**Yale University** — *New Haven, CT*
Doctoral Research; Graduate Research Assistant — *Sep. 2021–Present*

- **Quantum memory systems.** Designed two-dimensional QRAM layouts and parallel-query scheduling (MICRO 2023; ASPLOS 2025), with Python simulators for qubit reuse and error/resource tradeoffs.
- **Magic-state factories.** Co-developed a Python compiler for bivariate bicycle codes with Kun Liu. Our ISCA 2026 work with IBM Quantum compresses the 49-to-1 distillation protocol from 13 to 7 logical qubits. Code and data.
- **Structured fault tolerance.** Developed QRAM routing-state parity checks and one-fault-tolerant controlled-SWAP gates under erasure noise (2026 preprint); contributed to qLDPC injection, routing, and heterogeneous-code architectures.
- **Current directions.** Algorithm-driven fault-tolerant architecture and full-stack qLDPC resource estimation for quantum chemical simulation, including coherent data access and magic-state delivery.

**Pacific Northwest National Laboratory** — *Richland, WA*
Graduate Student Intern; Mentor: Dr. Ang Li — *Jun.–Aug. 2025*

- Worked on the OQRAM project.

**University of Chicago** — *Chicago, IL*
Visiting Ph.D. Student; Host: Prof. Liang Jiang — *Jan.–May 2024*

- Worked on Fat-Tree QRAM (ASPLOS 2025) and initiated subsequent qLDPC research projects.

## Publications

### Peer-Reviewed Publications and Accepted Papers

**Shifan Xu**, Kun Liu, Patrick Rall, Zhiyang He, and Yongshan Ding. "Distilling Magic States in the Bicycle Architecture." In Proceedings of the 2026 ACM/IEEE 53rd Annual International Symposium on Computer Architecture (**ISCA**), pp. 2081–2095. 2026. DOI: 10.1109/ISCA66397.2026.00148 · arXiv:2602.20546.

Kun Liu, Takahiro Tsunoda, Sophia H. Xue, Evan McKinney, Zeyuan Zhou, **Shifan Xu**, Robert J. Schoelkopf, and Yongshan Ding. "Efficient Routing of Quantum LDPC Codes on Programmable 2D Toric Architectures." IEEE International Conference on Quantum Computing and Engineering (**QCE**), 2026. arXiv:2604.18714.

Adrian Harkness, Shuwen Kan, Chenxu Liu, Meng Wang, John M. Martyn, **Shifan Xu**, Diana Chamaki, Ethan Decker, Ying Mao, Luis F. Zuluaga, Tamás Terlaky, Ang Li, and Samuel Stein. "FTCircuitBench: A Benchmark Suite for Fault-Tolerant Quantum Compilation and Architecture." ACM Transactions on Quantum Computing (**TQC**), 2026. arXiv:2601.03185.

**Shifan Xu**, Alvin Lu, and Yongshan Ding. "Fat-Tree QRAM: A High-Bandwidth Shared Quantum Random Access Memory for Parallel Queries." Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), Volume 2. 2025. DOI: 10.1145/3676641.3716256 · arXiv:2502.06767.

Samuel Stein, **Shifan Xu**, Andrew W. Cross, Theodore J. Yoder, Ali Javadi-Abhari, Chenxu Liu, Kun Liu, Zeyuan Zhou, Charles Guinn, Yufei Ding, Yongshan Ding, and Ang Li. "HetEC: Architectures for Heterogeneous Quantum Error Correction Codes." Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS**), Volume 2. 2025. DOI: 10.1145/3676641.3716001 · arXiv:2411.03202.

**Shifan Xu**, Connor T. Hann, Ben Foxman, Steven M. Girvin, and Yongshan Ding. "Systems Architecture for Quantum Random Access Memory." In Proceedings of the 56th Annual IEEE/ACM International Symposium on Microarchitecture (**MICRO**), pp. 526–538. 2023. DOI: 10.1145/3613424.3614270 · arXiv:2306.03242.

### Preprints

**Shifan Xu**, Connor T. Hann, Nathan Wiebe, Steven M. Girvin, and Yongshan Ding. "Error-Corrected QRAM Queries with One-Fault-Tolerant CSWAP Gates." arXiv preprint arXiv:2610.17185 (2026).

**Shifan Xu**[*], Yizhuo Tan[*], Yongshan Ding, and Jakub Szefer. "OQRAM: Oblivious Quantum Random Access Memory for Securing Delegated Quantum Queries." arXiv preprint arXiv:2607.27171 (2026). [*]Co-first authors.

Dantong Li, **Shifan Xu**, and Yongshan Ding. "Asymptotically Optimal Depth Fermionic Permutation on 2D Grid Quantum Architecture without Ancillas." arXiv preprint arXiv:2605.26041 (2026).

Kun Liu, **Shifan Xu**, Tomas Jochym-O'Connor, Zhiyang He, Shraddha Singh, and Yongshan Ding. "In-Situ Simultaneous Magic State Injection on Arbitrary CSS qLDPC Codes." arXiv preprint arXiv:2604.05126 (2026).

D. K. Weiss, **Shifan Xu**, Shruti Puri, Yongshan Ding, and Steven M. Girvin. "Faulty towers: recovering a functioning quantum random access memory in the presence of defective routers." arXiv preprint arXiv:2411.15612 (2024).

## Research Software & Tutorials

**PennyLane / Xanadu collaboration.** Developed the QRAM implementations and Python tutorial in collaboration with PennyLane, comparing select-only, bucket-brigade, and hybrid constructions through circuit simulation and resource analysis. Source code.

**Shifan Xu**, David Ren, Marcus Edwards, and Yongshan Ding. *Intro to quantum random access memory (QRAM)*. PennyLane Demos, August 30, 2026. Tutorial and code.

**Simhybridqram.** Developed a Python simulator for the MICRO 2023 QRAM architecture, including qubit reuse, error-scaling analysis, and resource tradeoff calculations. Source code.

## Technical Skills

**Programming languages:** Python, C.
**Research tools:** Fault-tolerant circuit compilation; quantum circuit simulation; QRAM simulators; resource estimation with PennyLane.

## Awards & Fellowships

2020 **ZhengGang Fellowship**, Nanjing University
2019 **People Scholarship**, Nanjing University
2019 **Outstanding Prize in Taiwan College Physics Experiment Tournament (TCPT)**, National Tsing Hua University
2018 **Outstanding Prize in Top-notch Talent Plan**, Nanjing University

## Teaching Experience

### Teaching Assistantships

**Yale University** *New Haven, CT, USA*
Graduate Teaching Assistant *Fall 2026*

- CPSC 4310/5310: Computer Music: Algorithmic and Heuristic Composition
- Instructor: Scott Petersen

**Yale University** *New Haven, CT, USA*
Graduate Teaching Assistant (two offerings) *2022 – 2023*

- CPSC 4470/5470: Introduction to Quantum Computing
- Instructor: Prof. Yongshan Ding
- Contributed to homework design with the teaching team and led multiple pre-exam review sessions.

### Guest Lectures

**Mar. 2026.** *Introduction to Quantum Query Architecture*. CPSC 4490/5490: Quantum Information Systems, Yale University. Host: Prof. Yongshan Ding.

**Feb. 2025.** *Introduction to Quantum Query Architecture*. CSCI 4965/6965: Quantum Computer System, Rensselaer Polytechnic Institute. Host: Prof. Zhiding Liang.

### Teaching Development

**Yale Certificate of College Teaching Preparation.** In progress; expected December 2026. Completed the workshops and learning communities.

## Mentoring

**Alvin Lu** *New Haven, CT, USA*
Undergraduate Student, Yale University *2024 – Present*

- Guided research from QRAM fundamentals to independent architectural exploration, leading to our *Fat-Tree QRAM* paper at ASPLOS 2025, presented by Alvin.

**Università del Piemonte Orientale** *Italy (remote)*
Informal Bachelor's Thesis Guidance *Jun.–Sep. 2026*

- Helped with Qiskit implementation.

## Presentations

### Conference Talks

**Jul. 2026.** *Distilling Magic States in the Bicycle Architecture*. 53rd ACM/IEEE International Symposium on Computer Architecture (ISCA).

**2023.** *Systems Architecture for Quantum Random Access Memory*. 56th Annual IEEE/ACM International Symposium on Microarchitecture (MICRO), Toronto, Canada.

### Invited Talks

**May 2026.** *Error-Corrected QRAM Queries with One-Fault-Tolerant CSWAP Gates*. AFOSR QRAM MURI closeout meeting (virtual). Invited by Prof. Steven M. Girvin.

**Apr. 2026.** *Distilling Magic States in the Bicycle Architecture*. Invited talk hosted by the Xanadu team.

**Spring 2024.** *Architectural Improvements in Quantum Random Access Memory*. Invited talk at Prof. Liang Jiang's Group Meeting, University of Chicago.

**Oct. 2023.** *Systems Architecture for Quantum Random Access Memory*. Invited talk at QAISG's Quantum Machine Learning Seminar, Centre for Quantum Technologies, National University of Singapore.

## Professional Service

### Program Committee

**2024.** Quantum Computing Applications and Systems Workshop, co-located with the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD). New Jersey, USA.

### Journal Refereeing

Physical Review Research
Physical Review A (**PRA**)
PRX Quantum
Physical Review Letters (**PRL**)
Physical Review X (**PRX**)
IEEE Transactions on Parallel and Distributed Systems (**TPDS**)
ACM Transactions on Design Automation of Electronic Systems (**TODAES**)
ACM Transactions on Quantum Computing (**TQC**)

### Conference Reviewing

Quantum Information Processing (**QIP 2026**)

## Earlier Research Experience

**2021. Nanjing University, Physics Department**, Nanjing, China. Graduation thesis: "Analysis of Background Noise in $p\bar{p}$ Collision Experiments Based on the K-Nearest Neighbors (KNN) Algorithm." Advisor: Dr. Tianjue Min.

**2020. University of California, San Diego, Department of NanoEngineering** (remote). Circuit Design for Flexible Ultrasonic Array. Advisor: Dr. Sheng Xu.

**2019–2021. Nanjing University, Engineering Department**, Nanjing, China. Preparation and Design of Auxetic Metamaterials. Advisors: Dr. Xin Li and Dr. Minghui Lu.

**2018–2019. Nanjing University, Physics Department**, Nanjing, China. Metal Powder Spectra Analysis. Supervisor: Prof. Sihui Wang.